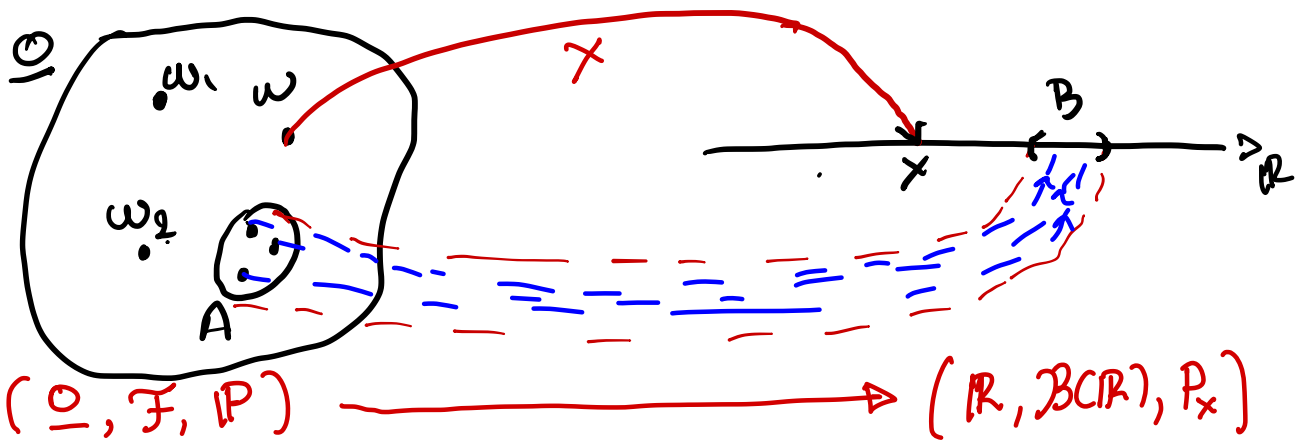
1.4 Introduction to Random Variables

Random Variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a measurable space of real numbers. A function $X : \Omega \to \mathbb{R}$ is called a ***random variable (R.V.)*** *if for every Borel set $B \in \mathcal{B}(\mathbb{R})$, the preimage of B under X is in $\mathcal{F}$, that is,*

$$\textit{Pre-image}(B) = X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} = \{X \in B\} = A \in \mathcal{F}.$$



Induced Probability

The probability measure of $(\mathbb{R}, \mathcal{B})$ is

$$P_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B) = \mathbb{P}(A)$$

P_X is called induced probability or distribution of X. (In measure theory, this is called pushed-forward measure)

So, a r.v. takes us from the probability space of the experiment $(\Omega, \mathcal{F}, \mathbb{P})$ to the probability space $(\mathbb{R}, \mathcal{B}, P_X)$, which has all the information associated with the r.v. X.

Comment: X is neither a variable nor random. It is a well-defined function. The correct term would be $\mathcal{F}$-measurable function that is used in measure theory

A Random Variable partitions the sample space

Toss a coin three times

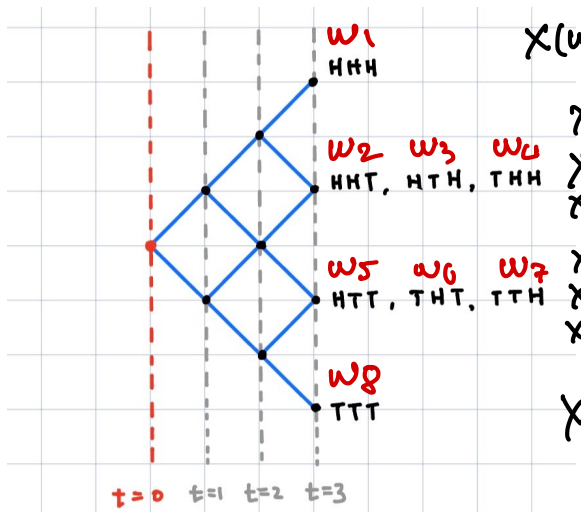
(Random walk)

R.V.

$X : \Omega \to \mathbb{R}$ s.t.

each H is $+1$

each T $= -1$



$X(\omega_1) = 3$

$\left.\begin{matrix} X(\omega_2) \\ X(\omega_3) \\ X(\omega_4) \end{matrix}\right\} = 1$

$\left.\begin{matrix} X(\omega_5) \\ X(\omega_6) \\ X(\omega_7) \end{matrix}\right\} = -1$

$X(\omega_8) = -3$

$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8\}$

$\mathcal{F} = \mathcal{P}(\Omega)$

$\mathbb{P}(\{\omega_i\}) = 1/8$

$(\Omega, \mathcal{F}, \mathbb{P})$

Define

$A_1 = \{\omega_1\}$

$A_2 = \{\omega_2, \omega_3, \omega_4\}$

$A_3 = \{\omega_5, \omega_6, \omega_7\}$

$A_4 = \{\omega_8\}$

$$X(\omega) = x \in \{-3, -1, 1, 3\}$$

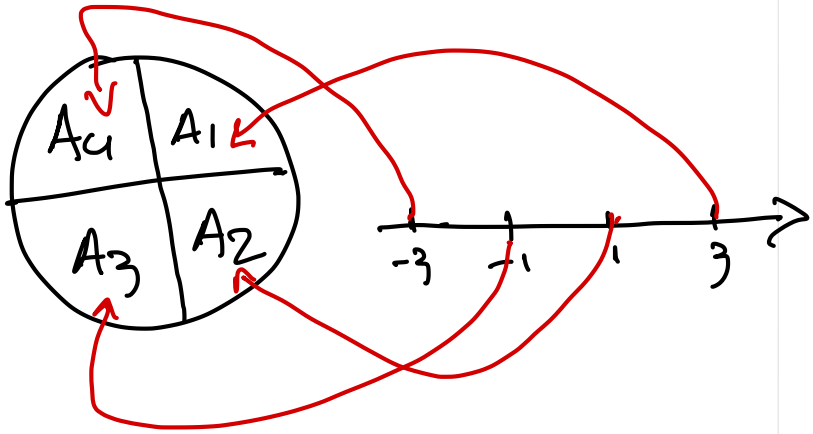
Note that

$X^{-1}(+3) = A_1$

$X^{-1}(+1) = A_2$

$X^{-1}(-1) = A_3$

$X^{-1}(-3) = A_4$



Comments

- A_i are disjoints and $\bigcup A_i = \Omega$
- This means that X partitions Ω
- $\forall \omega \in A_i$, $X(\omega) =$ constant

$\sigma-$Algebra generated by a random variable

The $\sigma-$algebra generated by a random variable X, $\sigma(X)$, is the smallest $\sigma-$algebra on Ω that contains all the sets $X^{-1}(B)$, $B \in \mathcal{B}$, i.e.,

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}\}$$

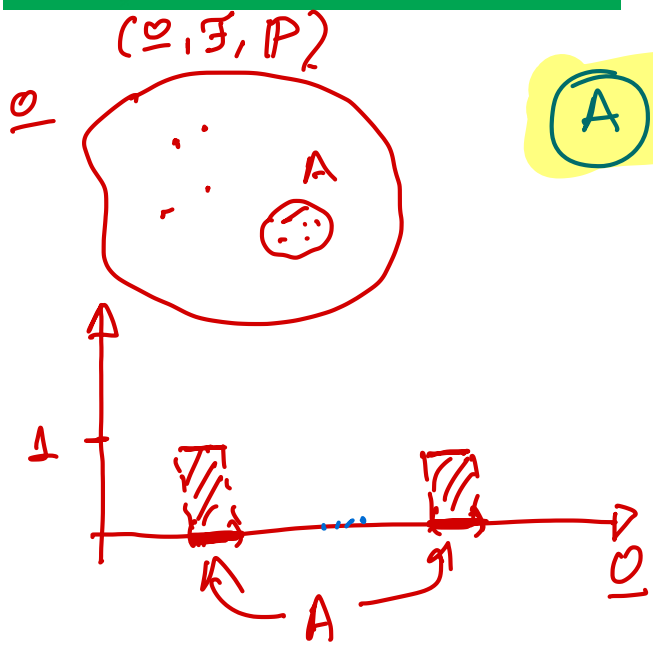
Clearly, X is $\sigma(X)$-measurable.

Equivalently, we can say that

$$\sigma(X) = \sigma(\mathcal{U}),$$

where $\mathcal{U} = \{A_1, A_2, A_3, A_4\}$

Integration of Random Variables



(A) Indicator function

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Assume $\mathbb{P}(A)$ is the "lenght" of A

$$\text{"Area"} = 1 \cdot \mathbb{P}(A) = \mathbb{P}(A)$$

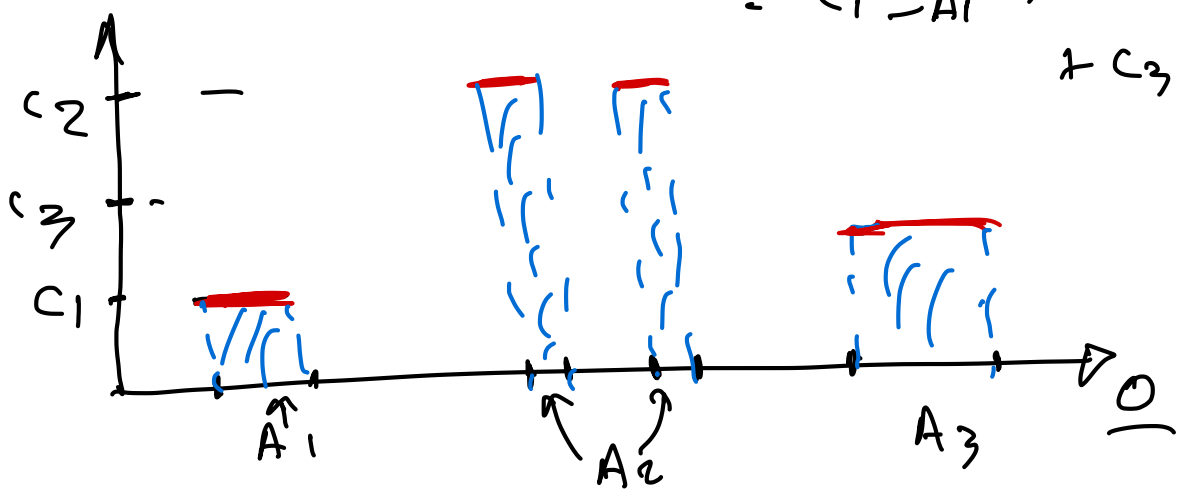
$$\| $$

$$\int_{\Omega} I_A \, d\mathbb{P}$$

(B) Simple functions

$\Phi : \Omega \to \mathbb{R}$ $\qquad \varphi(\omega) = \sum_{i=1}^{3} c_i I_{A_i}(\omega)$

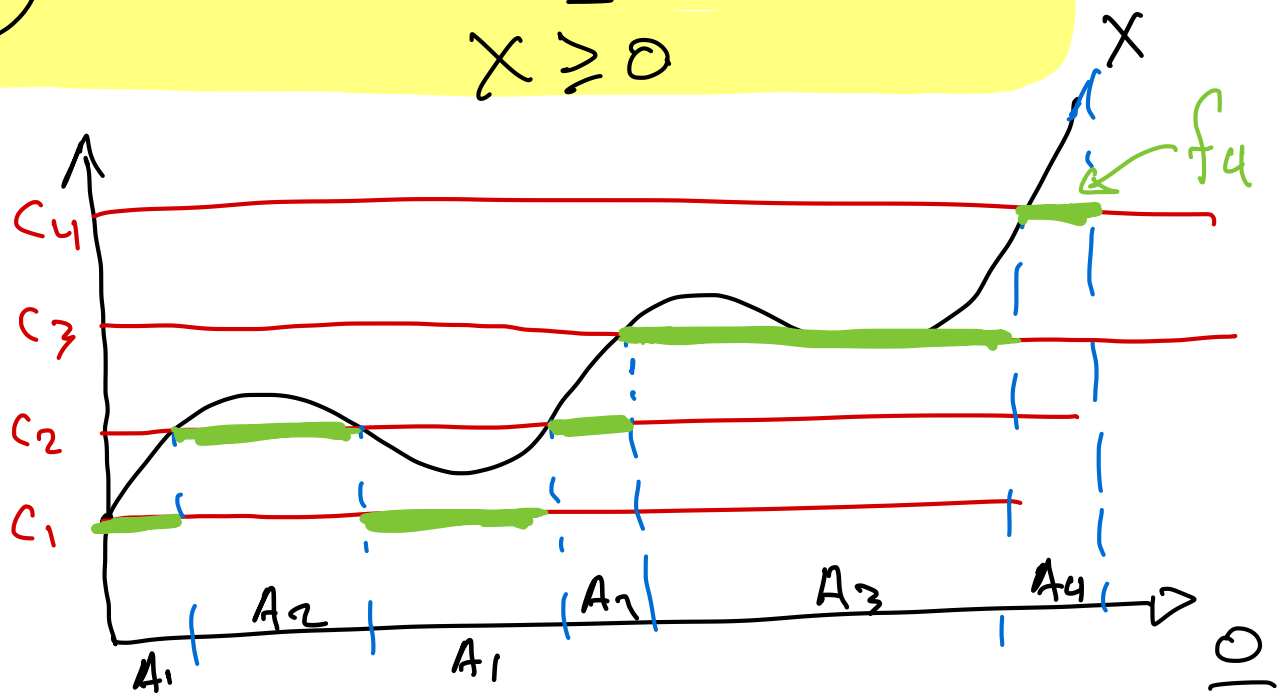
$$= c_1 I_{A_1}(\omega) + c_2 I_{A_2}(\omega) + c_3 I_{A_3}(\omega)$$



$$\int_{\Omega} \varphi \, d\mathbb{P} = c_1 \mathbb{P}(A_1) + c_2 \mathbb{P}(A_2) + c_3 \mathbb{P}(A_3)$$

$$= \sum_{i=1}^{3} c_i \mathbb{P}(A_i)$$

(c) A non-negative R.V. $X \geq 0$



$$X(\omega) \approx \sum_{i=1}^{4} c_i I_{A_i}(\omega) = f_4(\omega)$$

$$\int_\Omega X \, dP \approx \sum_{i=1}^{4} c_i P(A_i)$$

There exist a sequence of simple functions s.t.

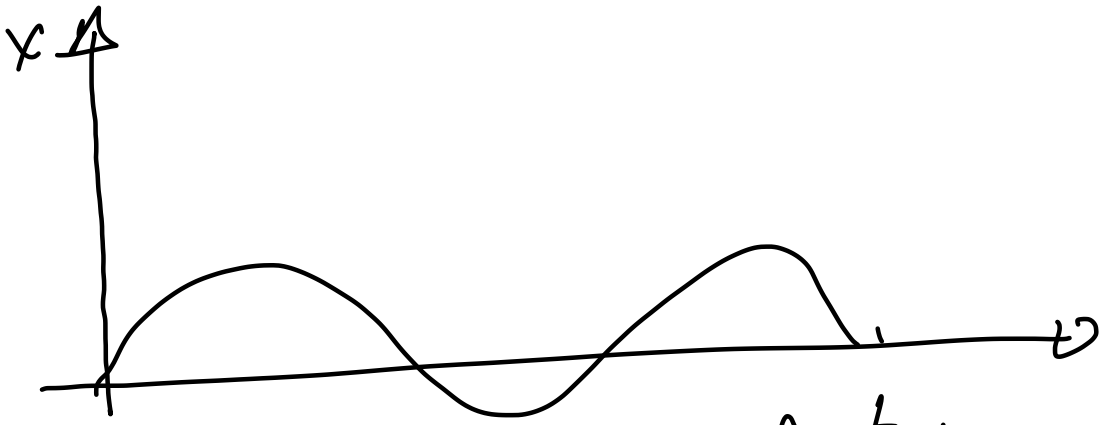
$$X(\omega) = \lim_{n \to \infty} f_n(\omega)$$

$$\int_\Omega X \, dP = c = \lim_{n \to \infty} \int f_n \, dP$$

Equivelently

$$\int_\Omega X \, dP = \sup \left\{ \int f \, dP, \quad 0 \leq f \leq X, \quad f \text{ simple function} \right\}$$

D) Not non-negative function.



non-negative $\begin{cases} X^+ = \sup\{X, 0\} & \checkmark \quad \int_\Omega X^+ \, d\mathbb{P} \\ X^- = \sup\{-X, 0\} & \checkmark \quad \int_\Omega X^- \, d\mathbb{P} \end{cases}$

$$\int_\Omega X \, d\mathbb{P} = \int_\Omega X^+ \, d\mathbb{P} - \int_\Omega X^- \, d\mathbb{P}$$

Both $\int_\Omega X^+ \, d\mathbb{P} < \infty$ and $\int_\Omega X^- \, d\mathbb{P} < \infty$

So $\int_\Omega X^+ \, d\mathbb{P} + \int_\Omega X^- \, d\mathbb{P} < \infty$

A R.V. is integrable if

$$\int_\Omega |X| \, d\mathbb{P} < \infty$$

Expectation

$$\mathbb{E}(X) = \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega)$$

$g : \mathbb{R} \to \mathbb{R}$.

$$E(g(X)) = \int_{\Omega} g(X(\omega))\, d\mathbb{P}(\omega)$$

Properties

$\mathbb{E}(cX) = c\,\mathbb{E}(X)$

$\mathbb{E}(X+Y) = \mathbb{E}(X) + E(Y)$ | linearity

- $X \geq 0 \quad E(X) \geq 0$ positivity
- $\mathbb{P}(A) = \int_{\Omega} I_A(\omega)\, d\mathbb{P}(\omega) = E(I_A)$

$(\Omega, \mathcal{F}, \mathbb{P})$

Monotone Convergence Theorem.

$\{f_n(x)\}_{n=1}^{\infty}$

1. $f_n(x) \leq f_{n+1}(x)$
2. $f_n(x) \to f(x)$, $n \to \infty$
3. $f_n(x) \geq 0$

$$\lim_{n\to\infty} \int_{\Omega} f_n \, d\mathbb{P} = \int_{\Omega} f \, d\mathbb{P}$$

Dominated Convergence theorem

1. $f_n(x) \to f(x)$, $n \to \infty$ (pointwise convergence)
2. $g(x)$ $\left(\int |g| \, d\mathbb{P} < \infty \right)$

$f_n(x) \leq g(x)$

$$\lim_{n\to\infty} \int_{\Omega} f_n \, d\mathbb{P} = \int_{\Omega} f \, d\mathbb{P}$$

$(\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$

$B \in \mathcal{B} \qquad \underline{P_X(B)} = \mathbb{P}(\underset{\in \mathcal{F}.}{X^{-}(B)})$

For instance $\underline{P_X((a,b])}$ is a set function

Cumulative distribution function

Define a point function $F_X : \mathbb{R} \to [0,1]$ *such that* $\quad \omega \in \Omega : X(\omega) \leq x.$

$$\underline{F_X(x)} = P_X((-\infty, x]) = \mathbb{P}(X \leq x).$$

The function F_X *is called* ***cumulative distribution function (CDF)****.*

$$P_X((a,b]) = \underbrace{F_X(b) - F_X(a)}_{\text{measur.}}$$

Properties of CDF

For any $x < y$*, we have*

$$\{X \leq x\} \subseteq \{X \leq y\} \quad \text{or} \quad \mathbb{P}(X \leq x) \leq \mathbb{P}(X \leq y).$$

Thus,

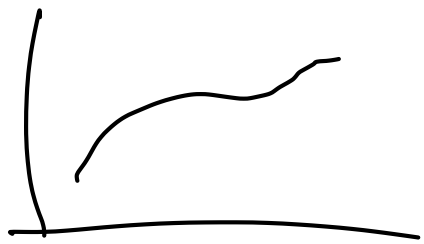
1. $\lim_{x \to -\infty} F_X(x) = 0$ *or* $\lim_{x \to \infty} F_X(x) = 1$,
2. $0 \leq F_X(x) \leq 1$,
3. $F_X(x)$ *is nondecreasing in* x,
4. $F_X(x)$ *need not be a continuous function, but it is always right continuous,*

$$\lim_{h \to 0^+} F_X(x+h) = F_X(x)$$

for all $x \in \mathbb{R}$.

Summarizing

$$\mathbb{E}(X) = \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega)$$



$$= \int_{\mathbb{R}} x \, dP_X(x)$$

$$= \int_{\mathbb{R}} x \, dF_X(x)$$

If $F_X(x)$ is absolutely continuous then

there is $\underbrace{f(x)}_{\text{PDF}} = \dfrac{dF_X}{dx}$ $\qquad F(x) = \int_{-\infty}^{x} f(x)\,dx.$

$$\int_{-\infty}^{\infty} dF_X = 1 = \int_{-\infty}^{\infty} f(x)\,dx, \qquad f(x) \geq 0$$

$$P_X([a,b]) = \int_a^b f(x)\,dx$$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f(x)\,dx.$$

$$E(g(X)) = \int_{\Omega} g(X(\omega))\, d\mathbb{P}(\omega)$$

$$= \int_{\mathbb{R}} g(x)\, d P_X(x)$$

$$= \int_{\mathbb{R}} g(x)\, d F_X(x)$$

$\rightarrow$

discrete R.V.

$$\sum_{\omega \in \Omega} g(X(\omega))\, \mathbb{P}(\omega)$$

$$\sum_{x_i} g(x_i)\, P_X(x_i)$$

Continuous R.V

$$\int_{-\infty}^{\infty} g(x)\, f(x)\, dx$$